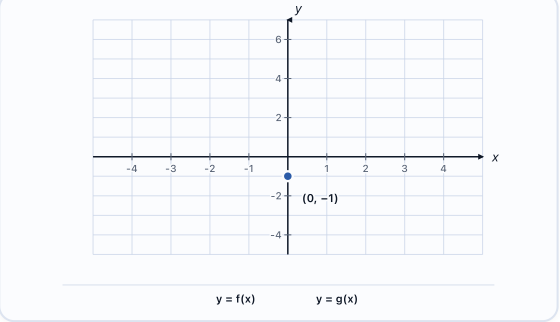
25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

GAT Ratios, Percentages, Work & Motion Problems Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 8x + 28$ in the domain of GAT Ratios, Percentages, Work & Motion Problems. By completing the square $f_1(x) = (x - h)^2 + k$, find the sum $k + f_1(6)$.



A 32
B 28
C 16
D 26

2 MCQ

A mathematical model in GAT Ratios, Percentages, Work & Motion Problems is defined by the cubic function $y_2(x) = 3x^3 - 4x^2 + 15x - 9$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 96
B 57
C 72
D 34

3 MCQ

In GAT Ratios, Percentages, Work & Motion Problems, two displacement vectors are given by $\mathbf{u}_3 = \langle 4, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.

A -20
B 80
C 172
D 60

4 MCQ

The linear growth profile shown in the diagram for GAT Ratios, Percentages, Work & Motion Problems follows an arithmetic progression with first term $a_1 = 5$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 705

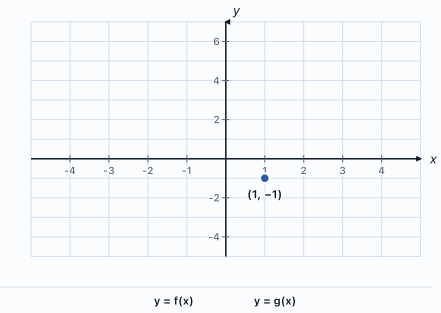
B 89

C 795

D 1335

5 MCQ

In GAT Ratios, Percentages, Work & Motion Problems, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 16x)\,\mathrm{d}x$.



A 104

B 80

C 96

D 41

6 MCQ

In GAT Ratios, Percentages, Work & Motion Problems, consider the quadratic function $f_6(x) = x^2 - 10x + 109$. Express $f_6(x)$ in the completed-square form $(x - h)^2 + k$ and compute the exact value of $k + f_6(12)$.

A 222

B 133

C 217

D 210

7 MCQ

The graph illustrates the cubic model $y_7(x) = 3x^3 - 5x^2 + 33x - 16$ studied in GAT Ratios, Percentages, Work & Motion Problems. Find the instantaneous gradient $\frac{\mathrm{d}y}{\mathrm{d}x}$ of the tangent to the curve at $x = 8$.

A 609

B 496

C 95

D 529

8 MCQ

In GAT Ratios, Percentages, Work & Motion Problems, two displacement vectors are given by $\mathbf{u}_8 = \langle 5, -7 \rangle$ and $\mathbf{v}_8 = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_8 \cdot \mathbf{v}_8 + \|\mathbf{u}_8\|^2$.

A. 79

B. 5

C. 74

D. 149

9 MCQ

A driver travels a $60\ \text{km}$ route from Riyadh to a research station at $30\ \text{km/h}$ and returns along the exact same route at $60\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
60	30

A. 13

B. 25

C. $45\ \text{km/h}$

D. $40\ \text{km/h}$

10 MCQ

A driver travels a $120\ \text{km}$ route from Riyadh to a research station at $40\ \text{km/h}$ and returns along the exact same route at $60\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
120	40

A. $48\ \text{km/h}$

B. $50\ \text{km/h}$

C. $53\ \text{km/h}$

D. 14

11 MCQ

A driver travels a $180\ \text{km}$ route from Riyadh to a research station at $60\ \text{km/h}$ and returns along the exact same route at $90\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
180	60

A. $70\ \text{km/h}$

B. $72\ \text{km/h}$

C. $75\ \text{km/h}$

D. $77\ \text{km/h}$

12 MCQ

A driver travels a $240\ \text{km}$ route from Riyadh to a research station at $60\ \text{km/h}$ and returns along the exact same route at $120\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
240	60

A $85\ \text{km/h}$

B $80\ \text{km/h}$

C $70\ \text{km/h}$

D $90\ \text{km/h}$

13 MCQ

A driver travels a $300\ \text{km}$ route from Riyadh to a research station at $20\ \text{km/h}$ and returns along the exact same route at $30\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
300	20

A $24\ \text{km/h}$

B $29\ \text{km/h}$

C $25\ \text{km/h}$

D $30\ \text{km/h}$

14 MCQ

A driver travels a $360\ \text{km}$ route from Riyadh to a research station at $50\ \text{km/h}$ and returns along the exact same route at $75\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
360	50

A $60\ \text{km/h}$

B 14

C $62.5\ \text{km/h}$

D $65\ \text{km/h}$

15 MCQ

A driver travels a $420\ \text{km}$ route from Riyadh to a research station at $80\ \text{km/h}$ and returns along the exact same route at $120\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
420	80

A $96\ \text{km/h}$

B $101\ \text{km/h}$

C $90\ \text{km/h}$

D $100\ \text{km/h}$

16 MCQ

A driver travels a $480\ \text{km}$ route from Riyadh to a research station at $45\ \text{km/h}$ and returns along the exact same route at $90\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
480	45

A $55\ \text{km/h}$

B $60\ \text{km/h}$

C $67.5\ \text{km/h}$

D $65\ \text{km/h}$

17 MCQ

A driver travels a $540\ \text{km}$ route from Riyadh to a research station at $60\ \text{km/h}$ and returns along the exact same route at $120\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
540	60

A $70\ \text{km/h}$

B $90\ \text{km/h}$

C $80\ \text{km/h}$

D $85\ \text{km/h}$

18 MCQ

A driver travels a $600\ \text{km}$ route from Riyadh to a research station at $80\ \text{km/h}$ and returns along the exact same route at $120\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
600	80

A $90\ \text{km/h}$

B $101\ \text{km/h}$

C $100\ \text{km/h}$

D $96\ \text{km/h}$

19 MCQ

A driver travels a $660\ \text{km}$ route from Riyadh to a research station at $120\ \text{km/h}$ and returns along the exact same route at $180\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
660	120

A $144\ \text{km/h}$

B $149\ \text{km/h}$

C $150\ \text{km/h}$

D $130\ \text{km/h}$

20 MCQ

A driver travels a $720\ \text{km}$ route from Riyadh to a research station at $120\ \text{km/h}$ and returns along the exact same route at $240\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
720	120

A $180\ \text{km/h}$

B $165\ \text{km/h}$

C $160\ \text{km/h}$

D $130\ \text{km/h}$

21 MCQ

A driver travels a $780\ \text{km}$ route from Riyadh to a research station at $40\ \text{km/h}$ and returns along the exact same route at $60\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
780	40

A $53\ \text{km/h}$

B 14

C $48\ \text{km/h}$

D $50\ \text{km/h}$

22 MCQ

A driver travels a $840\ \text{km}$ route from Riyadh to a research station at $100\ \text{km/h}$ and returns along the exact same route at $150\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
840	100

A $125\ \text{km/h}$

B $120\ \text{km/h}$

C 14

D $110\ \text{km/h}$

23 MCQ

A driver travels a $900\ \text{km}$ route from Riyadh to a research station at $160\ \text{km/h}$ and returns along the exact same route at $240\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
900	160

A $197\ \text{km/h}$

B $200\ \text{km/h}$

C $170\ \text{km/h}$

D $192\ \text{km/h}$

24 MCQ

A driver travels a $960\ \text{km}$ route from Riyadh to a research station at $90\ \text{km/h}$ and returns along the exact same route at $180\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
960	90

A $120\ \text{km/h}$

B $100\ \text{km/h}$

C $135\ \text{km/h}$

D $125\ \text{km/h}$

25 MCQ

A driver travels a $1020\ \text{km}$ route from Riyadh to a research station at $90\ \text{km/h}$ and returns along the exact same route at $180\ \text{km/h}$. What is the average speed for the entire round trip?

Given Motion Parameters

Parameter 1	Parameter 2
1020	90

A $100\ \text{km/h}$

B $135\ \text{km/h}$

C $120\ \text{km/h}$

D $125\ \text{km/h}$